Apocrine-Eccrine Hybrid Glands and Apocrine Sweat Glands

Apocrine-Eccrine Hybrid Glands (AEGs)

The relative frequency of apocrine-eccrine hybrid glands (AEGs) varies widely among adults. In individuals with axillary hyperhidrosis, up to 50% of axillary sweat glands—comprising eccrine, apocrine, and hybrid types—are of the apocrine type. In contrast, individuals without axillary hyperhidrosis show a much lower proportion of AEGs.

Like eccrine glands, AEGs possess a long duct that opens directly onto the skin surface, differing from true apocrine glands, whose ducts open into hair follicles. The secretory tubule of the AEG consists of two segments: a narrow segment with a diameter less than 80 μm (similar to the eccrine secretory tubule) and a second segment that undergoes uniform or irregular dilatation, reaching up to 500 μm in diameter. These segments are functionally connected, as demonstrated by micropuncture studies.

The thick segment of the AEG closely resembles the secretory portion of the apocrine gland, although some columnar secretory cells contain abundant mitochondria. The thin segment is histologically indistinguishable from the eccrine secretory coil. Functionally, the AEG responds to both cholinergic and adrenergic stimulation. Its secretory rate can be up to 10 times greater than that of eccrine glands, primarily due to its larger size, suggesting a potential role in axillary hyperhidrosis.

Notably, the AEG responds equally well to epinephrine and methacholine, indicating dual receptor sensitivity. It has been proposed that the AEG may represent an eccrine gland that has undergone "apocrinization" under the influence of local growth factors. If this hypothesis is correct, similar apocrinization processes might occur elsewhere in the body where such factors are present.

Apocrine Sweat Glands

In addition to eccrine and apocrine-eccrine hybrid glands, true apocrine sweat glands are present in humans, predominantly located in the axillae and perineum. These glands become functionally active just before puberty, suggesting hormonal regulation during sexual maturation, although the specific hormones involved have not been definitively identified. Notably, gonadectomy in mature individuals does not impair apocrine gland function, indicating that while their development is hormone-dependent, the maintenance of their activity is not. Thus, if classified as secondary sex characteristics, it is their development—not ongoing function—that relies on sex hormones.

Functions

Apocrine glands have been associated with several proposed functions, including: - Production of odoriferous substances that may act as sexual attractants - Territorial marking - Warning signals - Enhancement of frictional resistance and tactile sensitivity - Contribution to evaporative heat loss in some nonhuman species

Pheromone production by apocrine glands is well documented in many animal species. Given that human apocrine glands become active at puberty and produce odorous secretions after bacterial modification, it is plausible—though speculative—that they once served a sexual or social signaling role that may now be vestigial.

High levels of **15-lipoxygenase-2** are found in the secretory cells of human apocrine glands. This enzyme produces **15-hydroxyeicosatetraenoic acid (15-HETE)**, a ligand for the nuclear receptor **peroxisome proliferator-activated receptor-gamma (PPAR-γ)**, which may function as a signaling molecule involved in secretion or cellular differentiation.

In nonhuman furred primates, apocrine glands may also contribute to thermoregulation.

Composition of Secretion

Human apocrine sweat is initially milky, viscous, and odorless when secreted. Odor develops only after bacterial degradation on the skin. The precise chemical composition of human apocrine secretion remains poorly understood, primarily due to challenges in collecting pure samples uncontaminated by eccrine sweat.

Moreover, because apocrine glands often share a ductal opening with sebaceous glands (forming a sudosebaceous canal), their secretions mix with sebum. This admixture likely accounts for the milky appearance of fluid collected via duct cannulation.

Mode of Secretion

Cannulation studies of the human apocrine sweat gland duct have revealed that secretion occurs in a pulsatile manner. This pulsatility is believed to result from contractions of the myoepithelial cells that surround the secretory epithelium, facilitating the expulsion of secretory product.